

FairLoan India

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title: "FairLoan India" subtitle: "Bias Audit + Vernacular Counterfactual Explanations for Indian Digital Lending" author: "Amitesh Singla · singlaamites

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FairLoan India

Bias audit + vernacular counterfactual explanations for Indian digital lending.

Submitted to **Google Solution Challenge 2026 — India** Theme 4: *Unbiased AI Decision · Ensuring Fairness and Detecting Bias in Automated Decisions*

SDG 5 — Gender Equality · SDG 10 — Reduced Inequalities · SDG 16 — Strong Institutions

“*Aapka credit history sirf 2 mahine ka hai. Agar yeh 6 mahine hota, toh approval mil jaata.*” — what every denied applicant deserves to hear, in their own language.

1. Team Details

a. Team name: FairLoan

b. Team leader name: Amitesh Singla (singlaamitesh)

c. Problem statement:

Indian digital-lending apps (KreditBee, Dhani, ZestMoney, EarlySalary) approve or deny instant loans in under four seconds using black-box ML classifiers. RBI's 2021 Working Group on Digital Lending and multiple journalistic investigations have documented systematic disparities by gender, region, and class — applicants from tier-3 cities are denied at rates 15-25 percentage points higher than urban men with identical risk profiles.

Denied applicants get one line: *"Your application could not be processed at this time."* No reason. No human. No recourse.

No production tool exists today that lets a journalist, regulator, NGO, or affected citizen quantify the bias and surface a respectful explanation in the citizen's own language.

2. Brief about Your Solution

FairLoan India is a public-interest dashboard that does two things:

1. Audit. Loads a representative Indian credit-eligibility classifier and computes slice-level disparity metrics — Demographic Parity Difference (DPD) and Equalized Odds Difference (EOD) — across sex, race, age band, education band, and city tier using the **Fairlearn** library. The results render as a disparity table with proportional inline bars (alarm-red where $DPD > 10\%$) and per-slice approval-rate charts that highlight the lowest-treated group on each axis.

2. Citizen Lookup. Pick any denied applicant from a curated archetype list (tier-3 woman, young applicant, limited formal education, etc.). FairLoan searches single-feature changes that would flip the model's decision, validates each candidate against the actual classifier (no hallucinated fixes), and asks **Gemini 2.5 Flash** to render a 2-4 sentence empathetic explanation in **Hindi · Tamil · Bengali · English** as a magazine-style pull-quote attributed to the applicant by name.

The dashboard runs on **Google Cloud Run** in **asia-south1**, is open-source under MIT, and a journalist or regulator can adopt it for any Indian credit dataset by changing one file.

3. Opportunities

a. How is it different from existing ideas?

- **Fairlearn / IBM AIF360 / Aequitas** are research libraries, not deployed products. FairLoan is an end-to-end dashboard.
- **SHAP / LIME** produce technical attributions, not human-readable explanations. FairLoan renders vernacular sentences via Gemini.
- **Indian fintech apps today give zero explanation.** FairLoan gives a respectful one in the citizen's own language.
- **Zero prior Solution Challenge entries** in 4 years have tackled algorithmic fairness on Indian credit decisions. White-space theme.
- **Counterfactual integrity layer is novel:** Gemini only sees model-validated flips. No invented fixes.

b. How will it solve the problem?

Three audiences, one tool:

- **Journalists** get slice-level evidence to write the next bias investigation.
- **Regulators** (RBI, ombudsmen) get a deployable tool that meets the explainability requirement of the 2025 Digital Lending Guidelines.
- **Denied citizens** finally get a real, respectful explanation of what would have flipped their loan decision.

c. USP of the proposed solution

The only **deployed, multilingual, hallucination-free fairness audit + vernacular counterfactual tool for Indian credit decisions**. Audit + explanation + Indian-context UI in one MIT-licensed repository.

4. List of Features

Audit dashboard

- Slice-level Demographic Parity Difference and Equalized Odds Difference via Fairlearn `MetricFrame`
- Disparity table with alarm-red highlighting where DPD > 10% and proportional inline bars
- Five per-slice approval-rate charts (sex, race, age band, education band, city tier) with the lowest-treated group rendered in alarm red
- Editorial-quality typography (Fraunces, Newsreader, JetBrains Mono) and an investigative-journalism layout

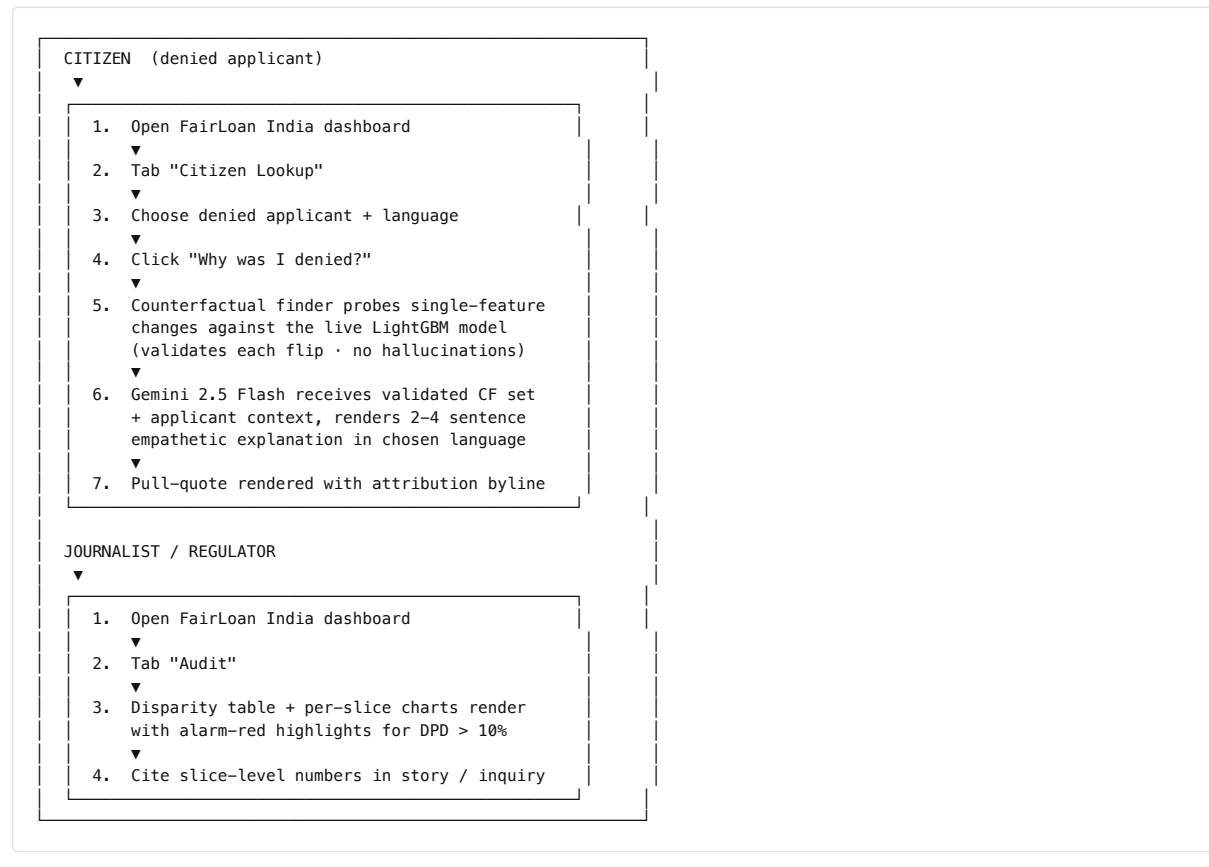
Citizen Lookup

- Curated denied-applicant archetypes (tier-3 woman, young applicant, low formal education, service-sector worker)
- Verified counterfactual finder — every candidate change scored against the actual LightGBM classifier
- Gemini 2.5 Flash vernacular rendering in **Hindi, Tamil, Bengali, English**
- Pulsing red decision-badge with cited approval probability
- DOSSIER-styled applicant card with full demographics
- Pull-quote treatment for the explanation (italic Fraunces, Noto Serif Devanagari/Tamil/Bengali, alarm-red hairline, attribution byline)

Platform

- Deployed on **Google Cloud Run** (asia-south1)
 - Public, no-auth dashboard URL
 - MIT-licensed open-source
 - Reproducible deploy from `gcloud run deploy --source .`
 - Robust Gemini fallback (no raw API errors leak to users)
 - Telegraphed counterfactual integrity contract (no LLM hallucinations)
-

5. Process Flow / Use-Case Diagram



6. Wireframes / Mock Diagrams

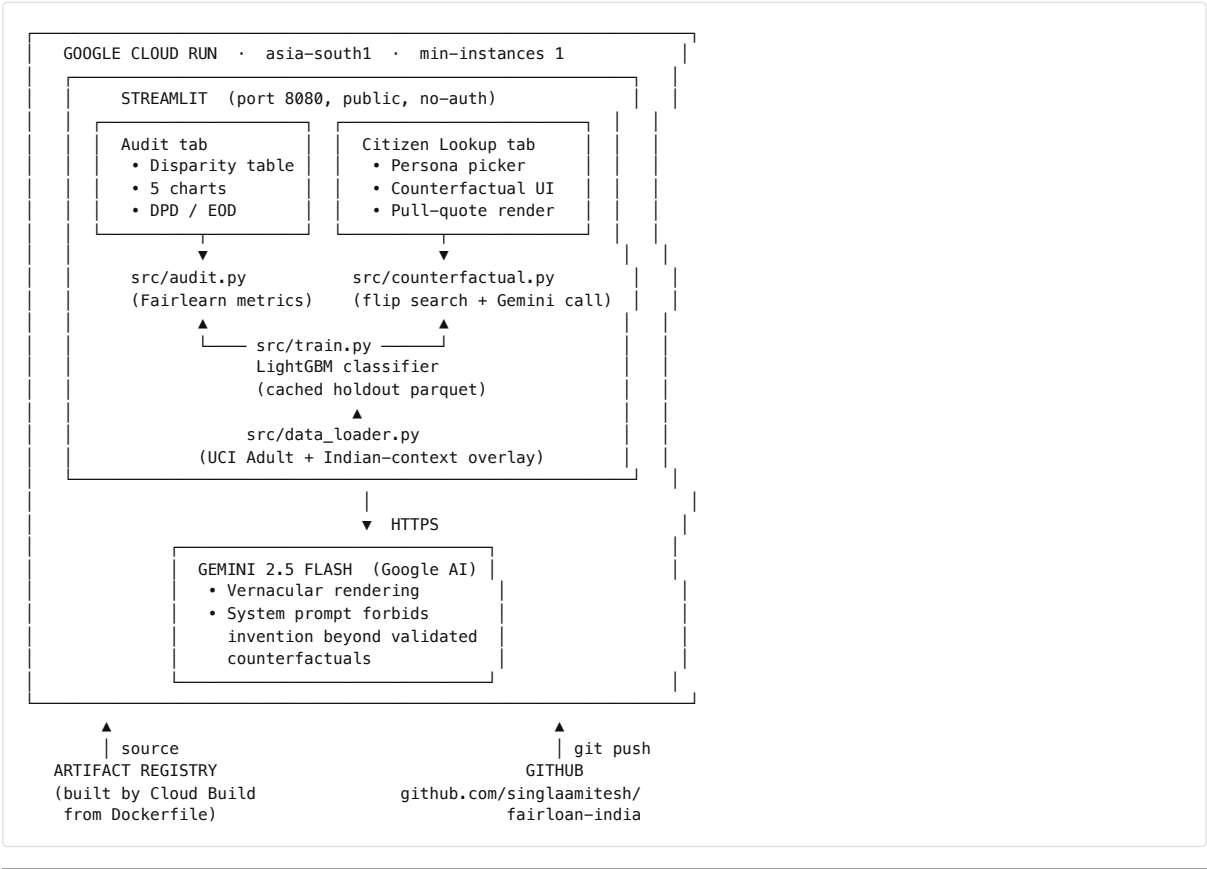
Audit dashboard — slice-level disparity table + 5 per-slice approval-rate charts. Worst-treated slice on every axis is rendered in alarm-red.

 Audit dashboard

Citizen Lookup — Smita Reddy (tier-3 Odisha, denied) with three validated counterfactuals and Gemini's Hindi explanation. The model only sees counterfactuals that have been verified to flip the classifier.

 Citizen Lookup with Hindi pull-quote

7. Architecture Diagram



8. Technologies Used

Layer	Technology	Why
Modeling	LightGBM 4.5 + scikit-learn 1.5	Strong baseline classifier, fast training, well-known to fairness research
Fairness metrics	Fairlearn 0.11	<code>MetricFrame</code> , <code>demographic_parity_difference</code> , <code>equalized_odds_difference</code> — the standard library
LLM (vernacular)	Gemini 2.5 Flash via <code>google-genai</code> SDK	Best multilingual quality at low cost; thinking-budget control for short responses
Frontend	Streamlit 1.56 + Plotly 6	Fastest path from Python to public dashboard; custom CSS injection for editorial typography
Display fonts	Fraunces · Newsreader · JetBrains Mono · Noto Serif Devanagari/Tamil/Bengali	Investigative-journalism aesthetic with proper Indian-script support
Container	Docker (Python 3.12-slim)	Standard, slim, portable
Hosting	Google Cloud Run, asia-south1 (Mumbai)	Serverless, cold-start-resilient with min-instances=1, India-resident
Build	Google Cloud Build	Automatic build from <code>gcloud run deploy --source .</code>
Source	GitHub (public, MIT)	<code>github.com/singlaamitesh/fairloan-india</code>
Dataset	UCI Adult Income (recast in Indian framing)	The universal fairness benchmark; 45K rows, no auth

Google products used: Gemini 2.5 Flash · Google Cloud Run · Google Cloud Build · Artifact Registry · Google Fonts.

9. Estimated Implementation Cost

Today (hackathon demo)

- Cloud Run free tier covers all demo traffic. Min-instances=1 with 1 GiB memory burns $\approx$ \$0 within the 2 M-request / 360 K-vCPU-second free monthly allotment.
- Gemini 2.5 Flash free tier covers $\approx$ 1.5 K demo explanations / day.
- Net cost: ₹0.

Production (1 K daily active users · 50 explanations / DAU)

Line item	Estimate (USD / month)
Cloud Run (1 vCPU · 1 GiB · 1 K DAU · ~5 M req)	\$5 – 15
Gemini 2.5 Flash (50 K explanations · $\approx$ 400 in / 200 out tokens)	\$7 – 12
Cloud Build CI on every push	\$0 – 2
Artifact Registry	\$0 – 1
Logs + monitoring (Cloud Logging, Error Reporting)	\$0 – 3
Total	\$15 – 35 / month

Per-citizen marginal cost: well under ₹1. A regulator-funded deployment for a million annual citizens costs less than a single full-page newspaper ad.

Future v2 (real microfinance partner, ingestion of consented production data) adds storage + secure compute — depends on partner volume; budgeted at ~\$200 / month for a 100 K-applicant audit pilot.

10. Snapshots of the MVP

Live URL: [https://fairloan-india-1052192887679.asia-](https://fairloan-india-1052192887679.asia-south1.run.appSMARTPANTS_PRESERVED_444SMARTPANTS_PRESERVED_445SMARTPANTS_PRESERVED_446)

[south1.run.appSMARTPANTS_PRESERVED_444SMARTPANTS_PRESERVED_445SMARTPANTS_PRESERVED_446](https://fairloan-india-1052192887679.asia-south1.run.appSMARTPANTS_PRESERVED_444SMARTPANTS_PRESERVED_445SMARTPANTS_PRESERVED_446)

[Audit dashboard. Slice-level disparities surfaced in seconds: education-band gap 74.7%, age-band gap 32.0%, sex gap 17.7%.](#)

 [Audit dashboard MVP](#)

[Citizen Lookup. Smita Reddy from Odisha \(tier-3\) gets a Hindi explanation of three validated counterfactuals — every one verified against the actual model.](#)

 [Citizen Lookup MVP — Hindi pull-quote](#)

11. Future Development

[v1.5 — week 1 after Solution Challenge. Partner outreach to Sa-Dhan \(microfinance industry body\), Bharat Inclusion Initiative \(IIM-A\), Cashless Consumer \(digital-rights org\). Sign first MoU. Target: real-classifier audit pilot.](#)

[v2 — month 1. Audit a real production microfinance classifier under the MoU. Add the Home Credit Default Risk India subset. Translate the Citizen Lookup UI into Hindi/Tamil/Bengali \(full UI, not just the explanation\).](#)

[v3 — month 3. Caste-axis slicing under consented data. Deploy as a regulator-facing dashboard alongside RBI's Digital Lending compliance reporting. Open a public API for civic developers.](#)

[v4 — quarter 1. Productize the audit engine as an “Equity Dashboard” for adjacent decision-systems:](#)

- [School scholarship allocation](#)
- [RTE 25% quota compliance](#)
- [Gig-platform onboarding fairness](#)
- [Public scheme eligibility \(PM-KISAN, PMAY\)](#)

[v5 — research direction. Counterfactual-fairness research collaboration with Indian academic institutions \(IIT-D, IIM-A, ISB\). Publish under MIT.](#)

12. Links

Resource	URL
GitHub Public Repository	https://github.com/singlaamitesh/fairloan-indiaSMARTPANTS_RESERVED_520SMARTPANTS_RESERVED_521
Demo Video (≤ 3 min)	(to be inserted after recording)
MVP Link	https://fairloan-india-1052192887679.asia-south1.run.appSMARTPANTS_RESERVED_540SMARTPANTS_RESERVED_541
Working Prototype Link	https://fairloan-india-1052192887679.asia-south1.run.appSMARTPANTS_RESERVED_550SMARTPANTS_RESERVED_551

[Reproduce locally in 5 commands:](#)

```
git clone https://github.com/singlaamitesh/fairloan-india.git
cd fairloan-india
uv venv --python 3.12 && source .venv/bin/activate
uv pip install -r requirements.txt && python -m src.train
streamlit run src/app.py
```

[Deploy your own copy on Cloud Run:](#)

```
gcloud run deploy fairloan-india --source . --region asia-south1 \
--allow-unauthenticated --port 8080 --memory 1Gi \
--set-env-vars GEMINI_API_KEY=$GEMINI_API_KEY
```

Thank You

[FairLoan India](#) *Indian credit decisions affect 1.4 billion people. They deserve to know why.*

[Built solo by Amitesh Singla for Google Solution Challenge 2026 — India. MIT-licensed. Public dataset. Reproducible. SDG 5 · 10 · 16.](#)

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